



as well as gaming benchmarks with plenty of charts to showcase differences. Coming to the rig, this is what we're running our benchmarks on.

Processor – AMD Ryzen 7 9800X3D

- CPU-Cooler – Noctua NH-D15
- RAM – 2x 32 GB Kingston FURY Beast 6000 MT/s
- SSD – Kingston KC3000 2TB
- PSU – Cooler Master V1200

The RTX 5080 delivers impressive performance across a range of demanding games. In *Cyberpunk 2077* at 1080p, it achieves a smooth 200+ FPS, dropping to a little over 140 FPS at 1440p and 71 FPS at 2160p. Enabling ray tracing impacts performance, with 105 FPS at 1080p and 35 FPS at 2160p. In *DOOM Eternal*, the card does quite well reaching 318 FPS at 1080p and maintaining strong frame rates at higher resolutions. *F1 24* also performs well, with 269 FPS at 1080p and a still-playable 158 FPS at 2160p. *Hogwarts Legacy* proves more demanding, but the RTX 5080 delivers 147 FPS at 1080p and 64 FPS at 2160p. Overall, the RTX 5080 offers excellent performance for high-refresh-rate gaming at 1080p and 1440p, with playable frame rates at 2160p, even with ray tracing enabled in some titles.

POWER CONSUMPTION AND THERMALS

At idle, the ASUS ROG Astral GeForce

RTX 5080 16GB GDDR7 OC Edition draws 22 watts, indicating that the card, despite its overclocked configuration, remains efficient when not under load. This helps keep system temperatures lower during general computing tasks and media consumption. However, power demands rise significantly under heavy load, with measurements peaking at 394 watts. This usage aligns with the card's high performance output, particularly during demanding gaming scenarios or complex workloads such as 3D rendering and AI-based calculations.

When installing this graphics card, it is crucial to consider both idle and maximum power usage. Many modern motherboards and power supplies include dedicated PEG connectors, designed for direct power delivery to the GPU. The recommended approach is to account for peak loads by selecting an appropriately sized power supply unit (PSU). NVIDIA suggests an 800-watt PSU for the RTX 5080, although the card's actual power usage can vary depending on system configuration and workloads. Even with an idle draw of 22 watts, overall system consumption should be evaluated to ensure adequate headroom under heavy load.

VERDICT

The ASUS ROG Astral GeForce RTX 5080 16GB GDDR7 OC Edition stands

as a strong contender for gamers and professionals looking for impressive performance, especially when paired with cutting-edge AI-driven features like DLSS4. While the card excels in rendering quality, performance improvements, and seamless gameplay at high resolutions. The heavy focus on Ray Tracing and Tensor cores, along with Multi-Frame Generation support, push the boundaries of what's possible in AI-assisted rendering, but the jury is still out on whether users will fully embrace this especially considering that the new 50-series cards are considerably more expensive. Furthermore, while power consumption is a concern, the card's performance and future-proof technology make it an attractive choice—although the steep MSRP remains a critical decision point for many buyers. Ultimately, the RTX 5080 offers a balanced mix of power and innovation, designed for those eager to step into the next generation of gaming technology.

—Mithun Mohandas

SPECIFICATIONS

CUDA Cores: 10752 | RT Cores: 84 | Tensor Cores: 336 | Base clock: 2295 MHz | Boost clock: 2617 MHz | VRAM: 16 GB | Memory interface: 256-bit | Process node: TSMC N4 | TDP: 360 W | Architecture: Blackwell

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